Resources ▼

blkdiag

Block diagonal matrix

Syntax

```
B = blkdiag(A1,...,AN)
```

Description

`B = blkdiag(A1,...,AN)` returns the block diagonal matrix created by aligning the input matrices `A1,...,AN` along the diagonal of `B`. example

Examples

collapse all

Diagonal of Three Matrices

Create a block diagonal matrix from three matrices of different sizes.

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Copy Command

```
A1 = ones(2,2);
A2 = 2*ones(3,2);
A3 = 3*ones(2,3);
B = blkdiag(A1,A2,A3)
```

Get

```
B = 7×7

     1     1     0     0     0     0     0
     1     1     0     0     0     0     0
     0     0     2     2     0     0     0
     0     0     2     2     0     0     0
     0     0     2     2     0     0     0
     0     0     0     0     3     3     3
     0     0     0     0     3     3     3
```

Input Arguments

collapse all

`A1,...,AN` — Input matrices
matrices

Input matrices, specified as a comma-separated list of matrices. The matrices can be either square or rectangular and can differ in size.

If any of the input matrices are sparse, then the output block diagonal matrix is also sparse.

More About

collapse all

Block Diagonal Matrix

A block diagonal matrix is a matrix whose diagonal contains blocks of smaller matrices, in contrast to a regular diagonal matrix with single elements along the diagonal. A block diagonal matrix takes on the following form, where $A1$, $A2$,..., AN are each matrices that can differ in size:

$$\begin{bmatrix} A1 & 0 & 0 & 0 \\ 0 & A2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & AN \end{bmatrix}$$

Extended Capabilities

collapse all

C/C++ Code Generation
Generate C and C++ code using MATLAB® Coder™.

GPU Code Generation
Generate CUDA® code for NVIDIA® GPUs using GPU Coder™.

Thread-Based Environment
Run code in the background using MATLAB® `backgroundPool` or accelerate code with Parallel Computing Toolbox™ `ThreadPool`.

The `blkdiag` function fully supports thread-based environments. For more information, see Run MATLAB Functions in Thread-Based Environment.

GPU Arrays
Accelerate code by running on a graphics processing unit (GPU) using Parallel Computing Toolbox™.

The `blkdiag` function fully supports GPU arrays. To run the function on a GPU, specify the input data as a `gpuArray` (Parallel Computing Toolbox). For more information, see Run MATLAB Functions on a GPU (Parallel Computing Toolbox).

Distributed Arrays
Partition large arrays across the combined memory of your cluster using Parallel Computing Toolbox™. (since R2024b)

The `blkdiag` function fully supports distributed arrays. For more information, see Run MATLAB Functions with Distributed Arrays (Parallel Computing Toolbox).

Version History

Introduced before R2006a

See Also

`diag`